

# COMPASS Initial Review and Alignment Memo

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**Name note:** Authored by **COMPASS** (agent renamed 2026-06-11; formerly “Master Planner”). See [COMPASS\\_AGENT\\_IDENTITY.md](#).

**Date:** 2026-06-11

**Author:** COMPASS (Cursor agent)

**Purpose:** Structured situational review after ingesting v6 guidance (VECTOR handoff), three domain agent reports, theory/manuscript corpus, dissertation proposal, Dakota v03 brief, and advisor memos.

**Status:** Near-term plan **largely locked** after Charles clarifications (2026-06-11). Advisor update: [20260611\\_Brief\\_for\\_Alex\\_Gates\\_brief.md](#) / full.

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## 1. Executive synthesis

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Charles Levine's dissertation program has **morphed substantially** since the August 2024 proposal (*Deciphering the Talent Mosaic* — prestige networks, unit eigenvector centrality, three Army-only chapters) while retaining core DNA: **Army officer careers as a closed talent system, peer and evaluator context matter, network-science and survival methods, and Barabási-style cross-domain universality ambition.**

The **current scientific center of gravity** (June 2026) is narrower and more defensible:

***Advancement under constrained distinction** — individuals compete in **local comparison pools** while **global distinction is finite**. Empirically, **advancement probability rises with LOO peer-pool quality through the middle of the distribution, then falls at the elite tier** (inverted-U). This pattern is **established in Army, replicated in NCAA basketball → NBA draft, and preliminary in R1 CS tenure.***

The project is transitioning from **empirical discovery** to **explanatory modeling** (Wang-style: stylized fact → minimal mechanism → non-obvious predictions). The **immediate convergence risk** is not lack of ideas — it is **parallel expansion** (525/UIC, network extensions, generative axis mismatches, tenure scrape expansion) displacing **minimal model closure, prediction list discipline, and manuscript draft.**

**VECTOR's handoff** is operationalized through [COMPASS\\_Initial\\_Guidance\\_v6.md](#) plus the theory corpus ([Vector\\_Master\\_Theory\\_and\\_Modeling\\_Notes\\_4.md](#), [Tier1\\_Narrative\\_Outline.md](#), Dakota v03 RTF, Wang model notes). There is no separate [VECTOR\\_report\\_to\\_COMPASS.md](#); v6 + 5-Manuscript folder is the VECTOR lane per Charles.

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## 2. Established vs preliminary — findings by domain

| Setting        | Agent | Outcome                            | Pool construct                                                         | Inverted-U                                           | Cox / survival                                                      | Manuscript-ready?                                                          |
|----------------|-------|------------------------------------|------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------------|
| 1 — Army       | CODA  | CPT→MAJ promotion vs attrition     | Senior-rater LOO pool TB ( <code>pool</code> minus <code>mean</code> ) | <b>Established</b> — CIF Q-bins + Cox quadratics     | <b>Yes</b> — Cell 11 CIF + Cell 12 cause-specific Cox               | <b>Strongest</b> — figures + HR tables exist; estimand language needs care |
| 2 — Basketball | SCOUT | NBA draft ( <code>Y_draft</code> ) | LOO teammate <code>poolq_loo</code>                                    | <b>Replicated</b> — ventile/binning draft rate       | <b>Partial</b> — Wang ladder (bins→LPM→logit); not time-to-event CR | <b>Strong empirical</b> ; generative claims must be qualified              |
| 3 — Academia   | PEER  | Tenure (Asst→Assoc) vs attrition   | LOO dept <code>poolq_loo_mean</code>                                   | <b>Preliminary</b> — stage 9, 18 bins, elite bin dip | <b>Wired, not reported</b> — Cells 10/10.5; Cell 12 pending         | <b>Preliminary figure + honest limitations</b> only                        |

### Confidence tags (COMPASS)

- **High confidence:** Army and basketball show the same *qualitative* inverted-U on LOO peer-quality axes (descriptive / binned / CIF layers).
- **Medium confidence:** Cox quadratic + interaction terms in Army support curvature on hazard scale; basketball LPM/logit yields turning point  $L^*$ .
- **Low-medium confidence:** Tenure stage-9 pattern (small N per bin, ~58% censoring, noisy OpenAlex linkage, unconditional bins).
- **Not yet supported:** Claim that **one minimal generative score** reproduces empirical inverted-U on  $L_Q$  **LOO axis** in simulation (SCOUT June finding: inverted-U on `team_mean`, mostly decreasing on  $L_Q$ ).

## 3. Theory / minimal model state (VECTOR framing)

### Working thesis (cross-domain)

From `Vector_Master_Theory_and_Modeling_Notes_4.md` and `Tier1_Narrative_Outline.md`:

Advancement depends on the interaction between **local peer quality** and **scarce distinction**. Stronger pools improve development/signaling up to a point; at the elite tier, **congestion, signal compression, and finite slots** dominate → inverted-U.

Tier 1 minimal objects

| Object      | Meaning                                   | Empirical proxy (basketball)                  |
|-------------|-------------------------------------------|-----------------------------------------------|
| $Y$         | Global advancement (distinction received) | <code>Y_draft</code>                          |
| $L_{ijt}$   | Amalgamated local environment             | Rolled into $Q$ first                         |
| $Q_{ijt}$   | LOO peer quality                          | <code>poolq_loo</code>                        |
| $\Lambda_t$ | Global distinction capacity               | Draft slots / promotion boards / tenure lines |
| $A_{ijt}$   | Own ability (control, not headline)       | PPM z                                         |

Minimal generative score (Alex / SCOUT):

$$S_i = A_i - \lambda \cdot L_{C,i}$$

where  $L_C$  = LOO viable-peer congestion. This is the **domain-agnostic minimal claim** — not the 539 notebook bundle.

Theory vs model discipline (Alex, Apr 30 2026)

`2026_0430_Paper7_feedback.md` and `2026_0507_Alex_Gates_Post_Meeting_Simulation_Memo.md` sharpen:

- **Theory** may include development, evaluator compression, sorting, prestige, networks.
- **Minimal model** should formalize the **smallest** set that reproduces the nonlinear pattern.
- **Quadratic in  $Q$**  is an **empirical diagnostic**, not the generative first principle.
- **Assortative pool formation + local comparison** is the generative story Alex emphasized (PD10–PD12).

Gaps vs empirical evidence

| Gap                                                                         | Severity                  | Notes                                                                                 |
|-----------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------|
| Generative inverted-U on same axis as 530 ( poolq_loo )                     | High                      | Same score, different Plot B axis → different shape (June 2026)                       |
| PEER Cell 12 Cox not archived                                               | High for formal Setting 3 | Stage 9 alone may suffice for dissertation <i>figure</i> but weakens parity with Army |
| Mechanism decomposition (B vs D) not separately identified                  | Medium                    | Tier 1 intentionally rolled-up; predictions list underdeveloped                       |
| Evans structural misclassification integrated in theory, not in minimal sim | Medium                    | Manuscript optional for v1                                                            |
| Network-science extensions (exposure vs comparison nets)                    | Low for v1                | Dakota §8 + Barabási brief — explicitly defer                                         |

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## 4. Cross-agent tensions and over-claim risks

### 4.1 Estimand conflation (highest manuscript risk)

| Tension                                      | Agents involved | Planner ruling                                                                                                                         |
|----------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------|
| CIF bar panels vs Cox HRs                    | CODA            | Keep layers distinct in prose; CIF = binned cumulative incidence; Cox = cause-specific hazards with censoring at competing event       |
| Cell 11 CIF estimator simplicity             | CODA            | Do not claim full Aalen–Johansen; last-snapshot bin assignment is implementation fact                                                  |
| Plot B axis: <code>team_mean</code> vs $L_Q$ | SCOUT / VECTOR  | <b>Not interchangeable</b> ; empirical paper leads with $L_Q$ ; generative POC may use <code>team_mean</code> — must say so explicitly |
| 539 vs 538D relationship                     | SCOUT           | Parallel bundles, not parent/child; minimal claim = <b>equation</b> , not notebook                                                     |
| Fine–Gray mentioned in Dakota/PEER           | PEER / VECTOR   | Code uses cause-specific Cox + empirical CIF; verify before manuscript claims Fine–Gray                                                |
| “168 schools” inference-ready                | PEER            | 168 <code>uni_slug</code> ≠ 168 equal-quality departments; distinguish coverage vs inference sample                                    |

### 4.2 Duplicated effort

| Duplication                                                            | Recommendation                                                                              |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Multiple Alex briefing outlines (v1–v4)                                | <b>Canonical:</b> <code>alex_gates_briefing_structure_outline_v4.md</code>                  |
| Multiple Scout status docs                                             | <b>Canonical:</b> <code>Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md</code> |
| <code>Vector_to_Scout_Tier1_Modeling_Direction.md</code> vs Alex spine | <b>Not sole canonical</b> per CODA Q11; use Alex spine + June Scout status                  |
| 535 vs 538 empirical paths                                             | Consolidate on <b>538/538D</b> unless Charles revives 535                                   |
| 537 generative sim                                                     | <b>Frozen</b> — benchmark only                                                              |

4.3 Hidden assumptions carrying weight

- 1. **LOO peer quality as sufficient statistic for local environment** — rolls development + congestion into one proxy (Tier 1 design choice).
- 2. **Three settings in one manuscript** — Charles confirmed 2026-06-08 (CODA report §10); dissertation bar = all three.
- 3. **Army proprietary data** — basketball/tenure are public-facing legs; Army figures may be illustrative without micro-data release.
- 4. **PPM+z as cross-domain performance harmonizer** — SCOUT still flags BPM/OBPM as open.
- 5. **Transcripts folder** — referenced in v6 but **not present in repo workspace**; advisor signal absorbed via memos ( `2026_0507` , `2026_0430` , `Alex_Tier1_Sequential_Model_Outline` , `TENURE_DATA_GAMEPLAN` § advisor).

4.4 Sequencing tensions — resolved vs open

| # | Issue                                                             | Status (2026-06-11)                                                                                                                                 |
|---|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Cell 12 Cox on tenure before Setting 3 prose                      | Resolved — <b>soft gate</b> : stage 9 + limitations OK for VECTOR draft; one Cell 12 Cox parallel before submission; Fine-Gray deferred             |
| 2 | SCOUT generative priority                                         | Resolved — <b>manuscript-first (Path A / Path II)</b> : honest layering + axis table; LOO generative match <b>deferred</b> ; nesting prose blocking |
| 3 | CODA parallel track (525/TB-stratify vs pool audit vs manuscript) | <b>Open</b> — Charles has not ranked; default <b>defer</b> per near-term plan                                                                       |
| 4 | OpenAlex confidence filter for tenure inference sample            | <b>Open</b> — HIGH only vs HIGH+MEDIUM vs MULTI with disclaimer                                                                                     |

5. Advisor alignment sketch (Alex Gates — preliminary)

Full `MENTAL_MODEL_OF_ALEX.md` deferred to Phase 1 embed in near-term plan; preliminary read:

What problem Alex believes we are solving

Show that *nonlinear advancement* emerges when *talent sorts into local pools* and *selection uses local comparison*, not raw ability alone — with a *minimal generative story*, not a quadratic baked in.

What evidence would convince Alex

- Cross-domain **empirical replication** (done for 2.5/3 settings).
- **Generative check**: ability-only fails; adding congestion/local comparison bends the curve.
- **Honest axis/estimand language** — no over-claim that one notebook “decomposes” another.
- **Wang ladder**: bins → transparent quadratic as **diagnostic** → logit  $L^*$  → generative mechanism.

What Alex would likely cut

- Network-science extensions before minimal model + predictions.
- Copying observed team means into generative assignment as “the model.”
- Presenting quadratic as **the** mechanism.
- Chasing perfect tenure scrape before end-to-end measurement on existing data.

What Alex would likely prioritize

- **PD11 threads:** (C) promotion score vs local rank tweaks → (A) soft-assignment pool formation → (B) mean × dispersion EDA.
- **End-to-end curves** with sample-loss accounting (PEER delivered this ethos).
- **Non-obvious predictions** (heterogeneity in draft-relevant tail; threshold congestion).

Risks Alex appears focused on

- Methodological overreach (causal claims, Fine–Gray without implementation).
- Generative/empirical axis mismatch presented as replication.
- Distraction from **paper** by 525 coding or URL scraping.

Laszlo Barabási (strategic): empirical replication + minimal mechanism + network-science **future** hooks  
( [barabasi\\_briefing\\_outline\\_v2.md](#) ). Lead with stylized fact, not closed proof.

6. Dissertation evolution — original vs current

| Dimension     | Aug 2024 proposal                                        | June 2026 program                                   |
|---------------|----------------------------------------------------------|-----------------------------------------------------|
| Title/framing | Talent mosaic, unit prestige, OCN eigenvector centrality | Advancement under constrained distinction           |
| Domains       | Army only (3 chapters)                                   | Army + basketball + academia                        |
| Core RQs      | Prestige units, peer effects, TB tradeoffs               | Inverted-U + minimal mechanism + predictions        |
| Methods       | OCN/OPN networks, survival                               | Survival + LOO pool metrics + Tier 1 generative lab |
| Committee     | Barabási chair, Gates, Vespignani, Dakota Murray         | Same; Dakota received v03 brief                     |

**Retained:** Army as anchor; Evans misclassification; competing risks; people-analytics audience; Wang/Barabási scientific style.

**Deprioritized for immediate manuscript:** Full OCN prestige validation (H1a–c), 525 UIC consistency as *paper* centerpiece (still valuable for Army depth / dissertation later).

## 7. Charles decisions (locked) and remaining open questions

Locked (COMPASS thread, 2026-06-11)

| # | Topic                     | Decision                                                                                                        |
|---|---------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1 | Hard deadline             | Summer–Fall 2026 core manuscript draft/submission                                                               |
| 2 | Tenure Cox gate           | Soft gate — see §4.4 #1                                                                                         |
| 3 | SCOUT generative priority | Manuscript-first — see §4.4 #2                                                                                  |
| — | Terminology               | LOO in any name for quantities excluding self                                                                   |
| — | Alex comms                | Plan update briefs ( <code>*_Brief_for_Alex_Gates_*.md</code> ) — Charles’s calls, not a decision questionnaire |

### Still open

- 4. CODA parallel track (§4.4 #3) — TB-stratify on AWS? pool-size audit before manuscript?
- 5. Transcripts location — `transcripts/*_Paper_directions*` not in workspace; memos sufficient?
- 6. Army data in planning docs — what can be named externally?
- 7. Stale file flags — any v6 paths superseded beyond agent reports?
- 8. Who owns Dakota RTF refresh when domain status shifts?
- 9. `PERF_METRIC` lock (PPM+z) vs BPM for cross-domain comparability?
- 10. OpenAlex confidence policy for tenure (§4.4 #4).

## 8. Draft agent question queue

| File                                                      | Rationale                                                                               |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>20260611_0826_COMPASS_to_VECTOR_questions.md</code> | Minimal model closure criteria; prediction prioritization; manuscript section ownership |
| <code>20260611_0826_COMPASS_to_SCOUT_questions.md</code>  | Generative $L_Q$ gap vs manuscript scope; axis figure requirements                      |
| <code>20260611_0826_COMPASS_to_PEER_questions.md</code>   | Setting 3 maturity target; Cox timeline; inference sample definition                    |
| <code>20260611_0826_COMPASS_to_CODA_questions.md</code>   | Army estimand language for manuscript; deferred TB-stratify / pool audit                |

Charles routes these to agents; COMPASS will not assume answers.

## 9. Provisional answer — shortest defensible path

*If the objective is to complete the minimal model, generate robust predictions, and produce a publishable manuscript, what is the most scientifically defensible path from today?*

Locked path (Charles, 2026-06-11):

```
LOCK CLAIM LANGUAGE (all agents)
↓
SCOUT: Manuscript exports + axis table + nesting chain for VECTOR §3
      (defer L00 generative bin-for-bin match; parallel north-star sprint if time)
↓
VECTOR: Draft Wang-structure manuscript – empirical triad → minimal model → predictions
↓
PEER: Cell 12 Cox in parallel (soft gate – not blocking first VECTOR draft)
↓
CODA: Manuscript figures + estimand-checked captions; 525/TB-stratify deferred
↓
PREDICTIONS: Lock 2 testable (heterogeneity + one dispersion/stratification candidate)
↓
SUBMIT Summer–Fall 2026; defer networks, Fine–Gray, full L00 generative match
```

Detail: `PROJECT_STATUS_AND_NEAR_TERM_PLAN.md` and `20260611_Brief_for_Alex_Gates_full.md`.

## 10. Review status

- §§1–4 situational understanding: Charles affirmed via clarification thread (2026-06-11).
- §9 path: Locked for near-term; open items in §7.4–10 remain.
- `PROJECT_STATUS_AND_NEAR_TERM_PLAN.md`: Active living doc; revise when agent answers land.
- Alex: `20260611_Brief_for_Alex_Gates_brief.md` / `_full.md` — plan update (Charles’s calls), not a decision memo.

End initial review memo.